WORKSHEET

# Chapter 6: Complex polynomials

**1** Identify the leading term and degree of each of the following complex polynomials.

**a** $p(z) = 4iz + 3i + 6iz^2 + iz^5 - z + 2z^4 - 3z^3 + 2z^5 + 4iz^3 - 2z^2 + 9iz^4$

**b** $r(z) = -11z^2 + 6iz + 2i + 8z^5 + 9z - 3z^7 + 4iz^2 + iz^5$

**c** $s(z) = 10iz + 4i + 3z^3 - 7z^2 + 8z + 5iz^3 + 2iz^2$

**d** $t(z) = 7iz^3 + 11iz^2 + 6iz + 5iz^4 - 2z^3 - 3z^6 + 15z^2 + 9z + 4i + 8iz^6$

**2** $m(z) = z^3 - 2z^2 + 3iz + 4 - 7i$. Find each of the following.

**a** $m(3)$

**b** $m(3i)$

**c** $m(-2)$

**d** $m(-2i)$

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**e** $m(1 - 3i)$

**f** $m(-2 + i)$

**3** $n(z) = 2z^4 + 4iz^3 + (3i - 5)z - 9 + 6i$. Find each of the following.

**a** $n(-3)$

**b** $n(4i)$

**c** $n(-2 + i)$

**d** $n(1 - 3i)$

**e** $n(3 + 4i)$

**f** $n(-3 - 2i)$

**4** $p(z) = (2 - i)z^2 + (-3 + 4i)z + 11 - 7i$. Find each of the following.

**a** $p(3)$

**b** $p(4i)$

**c** $p(5 + 2i)$

**d** $p(-3-2i)$

**e** $p(2-5i)$

**f** $p(-3-7i)$

**5** CAS $q(z) = (7-5i)z^4 + (-3-8i)z^3 - 12z + 9i$. Find each of the following.

**a** $q(6)$

**b** $q(-4i)$

**c** $q(3+7i)$

**d** $q(-6-i)$

**e** $q(6+i)$

**f** $q(8+5i)$

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**6** CAS $h(z) = (1 - 4i)z^3 - (3 + 2i)z^2 + (8 - 3i)z - 12 + 17i$. Find each of the following.

**a** $h(-2)$

**b** $h(5i)$

**c** $h(-1 + 4i)$

**d** $h(3 - 5i)$

**e** $h(3 + 5i)$

**f** $h(-2 - 3i)$

**7** CAS $r(z) = (1 - 2i)z^5 + (3 + 5i)z^3 - (8 - 3i)z^2 + 6z - 12 + 15i$. Find each of the following.

**a** $r(1 + 3i)$

**b** $r(-2 + 3i)$

**c** $r(5 - 2i)$

**d** $r(7 + 3i)$

**e** $r(-1 - 5i)$

**f** $r(-3 - 4i)$

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## Answers

**1** **a** $(2 + i)z^5$; 5

**b** $-3z^7$; 7

**c** $(3 + 5i)z^3$; 3

**d** $(-3 + 8i)z^6$; 6

**2** **a** $13 + 2i$

**b** $13 - 34i$

**c** $-12 - 13i$

**d** $18 + i$

**e** $3 + 26i$

**f** $-7 + 6i$

**3** **a** $168 - 111i$

**b** $747 - 14i$

**c** $-60 - 61i$

**d** $-21 + 112i$

**e** $-1266 - 1145i$

**f** $-42 + 283i$

**4** **a** $20 - 4i$

**b** $-37 - 3i$

**c** $50 + 26i$

**d** $50 + 6i$

**e** $-37 - 3i$

**f** $10 + 126i$

**5** **a** $8352 - 8199i$

**b** $2034 - 1415i$

**c** $-17\,974 - 19\,001i$

**d** $11\,577 + 2401i$

**e** $11\,957 - 1433i$

**f** $3895 + 66\,223i$

**6** **a** $-48 + 47i$

**b** $-422 - 18i$

**c** $-140 - 134i$

**d** $-253 + 872i$

**e** $-23 + 792i$

**f** $12 - 220i$

**7** **a** $344 - 867i$

**b** $-1171 + 18i$

**c** $-9224 - 1207i$

**d** $32\,804 + 47\,313i$

**e** $-6860 + 4385i$

**f** $6994 + 2873i$

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